

KEY DIMENSIONS - modelwheels.py TABLES				
ITEM (mm)	MAIN	NOSE	SOURCE	
Tyre (modelled)	8.50-18 Type III	17.5x5.25-6	0 owner decision 2026-09-26; 5 tyre makers / parts lists	
Part numbers	86906C-3 / 805-358-8	1706881 / 651-327-6	5 Goodgear / Michelin (parts listings)	
Ply / type / pressure	18 PR TL, 60 psi	8 PR TL, 60 psi	5 data book: VME placard, 556-82527	
Fro OD / section W	648 / 216	448 / 150	5 TRA 627-652 / 208-221; N photos 628-658	
Acta M = loaded R / defl.	279 / 41	287 / 15.5	5 main axia kept; N main 278-289, nose 288-210	
Contact patch / bleed R	268 / 189	139 / 78	E (free choice 312 / 163)	
Section H / aspect	193.6 / 6.89	146.5 / 6.92	0 (TRA 9.98 / 6.92)	
Head wheel / flange dia	254 / 265.2	152 / 156	5 TRA 138-28 / 6.25; 1 flange .85 / .75 in	
between Flanges / ledge	155.8 / 34.3	127 / 22.9	5 TRA 6.25 / 5.88 in; 1.35 / 6.99 in	
Crown R / tread width	288 / 164	128 / 135	E main; N nose / N	
Shoulder R / sidewall R	12 / 162	18 / 97	E / 0 (calved tangent)	
grooves: centres +- groove width x depth	36, 56 (2 pairs) 8 x 7	15, 44 (even) 6 x 5	N head-on photos 3696, 3988, 3981	
Rain centre / air / water	52 / 18 / 22	24 / 23 / 11	N width / E depth	
wheel	split hub, 2 halves	split hub, 2 halves	5 POW 7-4-11; Goodrich 3-1543 / 3-1581	
Tie bolts n on PCO	8 on 288	4 on 130	E hidden (nail); N PCO, E count (nose)	
Web face s (skin depth)	11 (split plane)	32 (38 deep)	E main (covered); N nose (split = hub 1/4 in seen)	
Valve / safety plug	8.77 / 1	R 66 / 3	N position; 5 POW	
Fuelite plug	3 (inboard half)	-	5 POW 7-4-11; E position	
Hub fairing tip / face / proud	298 / 264 / 28	-	N 3606, 3605, 3658, 3608, NIX	
fairing screws / hole	5 on 288 / 855 Ø77	-	N 5 at 72 deg	
Brake	8 Ø58.1; 3 REF; 1 steel	-	5 POW 7-4-18; kit 2-1824-1	
Brake discs	3 rotors / 2 stators	-	5 stators / E rotors	
Housing lobes / rotor dia	212 / 232	-	N lobe tips / E	
Brake stack	-189 - - 28	-	E (x = 8 downward)	
Acta dia / bore	52 / 45	48	5 gear.py / N open bore	
Fork arms (faces)	-	+83, 118	N two arms (head-on photos)	

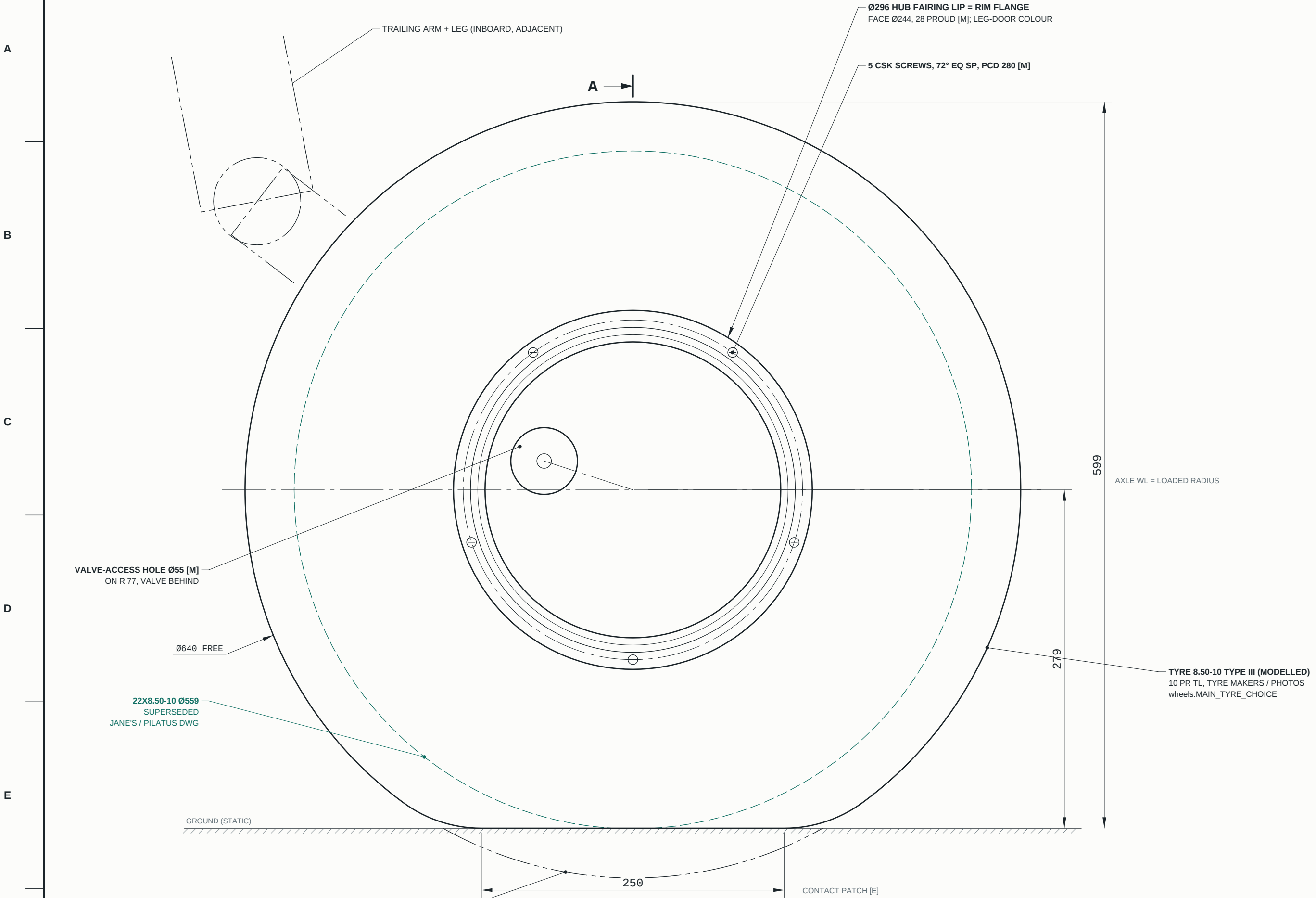
0 owner decision / proposed / sourced / N photo-referenced / E measured / Drawn / G gear.py / 3-D = main-referenced (used 3D)

PHOTO CHECK - DEVIATION CALL-OUTS (mm)				
ITEM	PHOTOS (M)	TABLE / MODEL	S	ACTION
Main tyre OD (modelled)	628-658	648	-	0 owner decision 2026-09-26
208x18-18 (upper-referenced)	-	659	-	0 MAIN TYRE CHOICE
3-D gear MAIN TYRE	-	648	-	0 From wheels.py
Main section width	214-224	216	-	0 -
Main loaded radius	279-288	axia 279	-	0 at flat draw
Tyre R / fairing lip R	2.15-2.21	2.16	-	0 -
retracted tyre depth	-25 (POW 1 in)	23.5	-	-1.9 28-38 (fit check 5)
Main grooves / W	8.54, 9.28	8.54, 9.28	-	0 3-D geometry
Render groove shader	-	off (geometry)	-	0 -
Nose tyre OD	37.5 in	445	-	0 15.5 Flat draw
Nose loaded radius	288-218	axia 297	-	0 3-D volu, crown block
Nose grooves / W	6.85, 8.28	8.89, 8.28	-	0 steel_dark
Nose fork arms	2	gear.py 2	-	0 3-D volu, crown block
Nose axle nut / lock plate	3-D both arms	3-D both arms	-	0 steel_dark
Nose tin-bolt PCO	dark hex + tear-drop	gear.py 2	-	0 3606 nose-hub zoom
Hub fairing dia / proud	-8.58 flange R	118	-	0 face 9 > W/2
Main pressure	298 / 26	298 / 26	-	0 gear.py / N open bore
Main brake	68 psi placard	gear.py info 60	-	0 -
	6-Lobe, bright cast	3-D: brake_housing	-	0 -

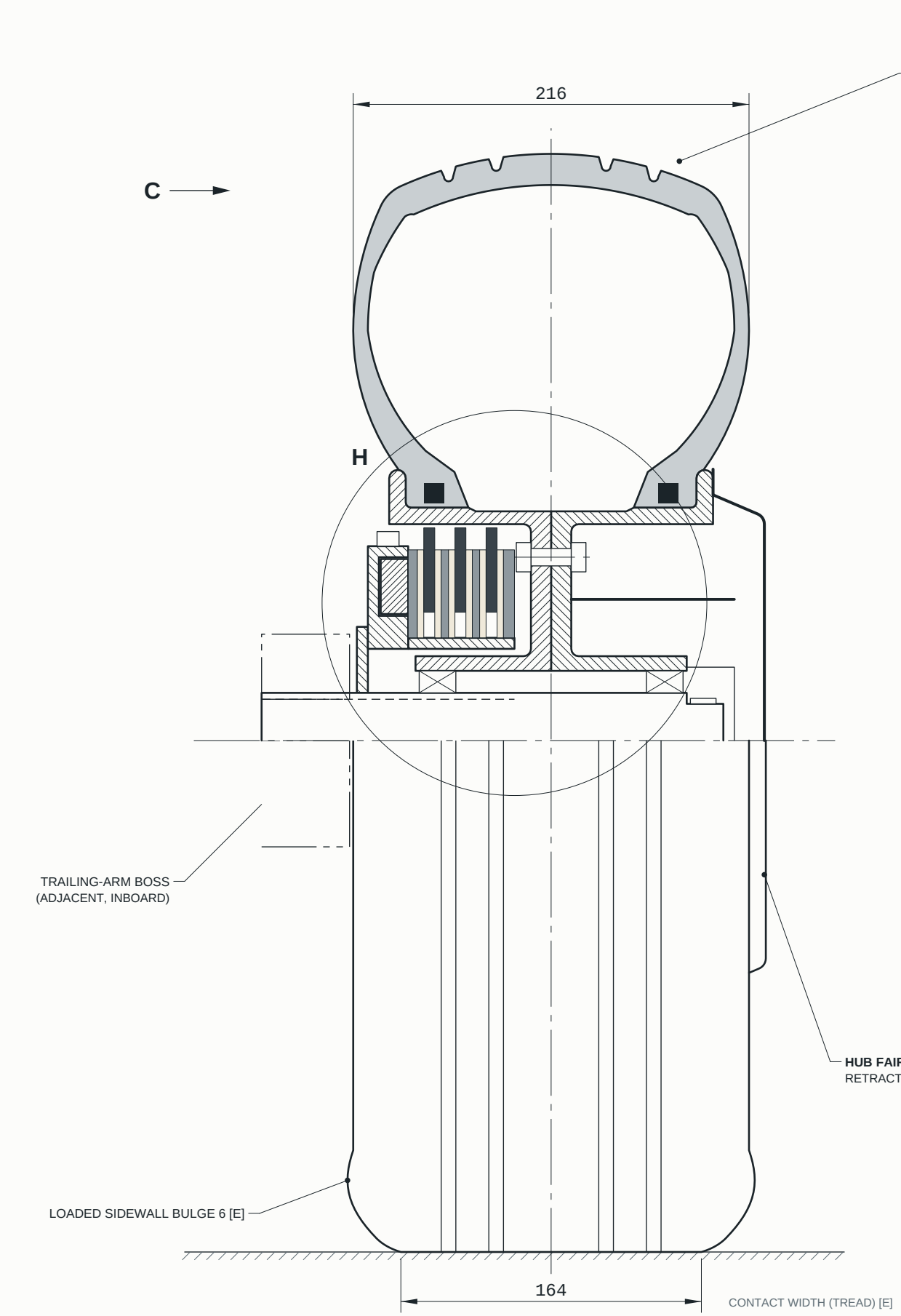
MATERIALS (modellassemble.py NAMES)				
PART	MAIN	NOSE	NOTE	
Tyre	tyre	tyre	black satin; NO lettering	
wheel halves	wheel	wheel	gloss light grey / white	
hub fairing	main_gear_door	-	leg-door colour (3808 blue)	
brake housing / discs	brake_housing / metal_dark	-	bright satin cast alu / dark steel	
Bolts, valve stem	cadmium	cadmium	yellow-chrome cadmium / brass	
fairing screws, valve cap	metal / black	-	rubber-sealed cap	
axia nut, hub cap	steel	steel_dark	dark steel nut + tear-drop lock plate (nose)	
bearing caps (nose hub ends)	chrome	chrome	polished seal rings	
arm / fork	gear_leg	gear_leg	as-patent	

LEGEND		SCALE 1:3	
	tyre (solid in section, metal in section (shaded))	0 0.05 0.1 0.15 0.2 0.25 0.3m	
	adjacent parts, free tyre, TRA env.		
	brake rotor / stator, living		
	hidden		
	centre line, pitch circle		
	22x8.50-10 tyre (superseated)	SCALE 1:2 0 0.025 0.05 0.075 0.1 0.125 0.15m	

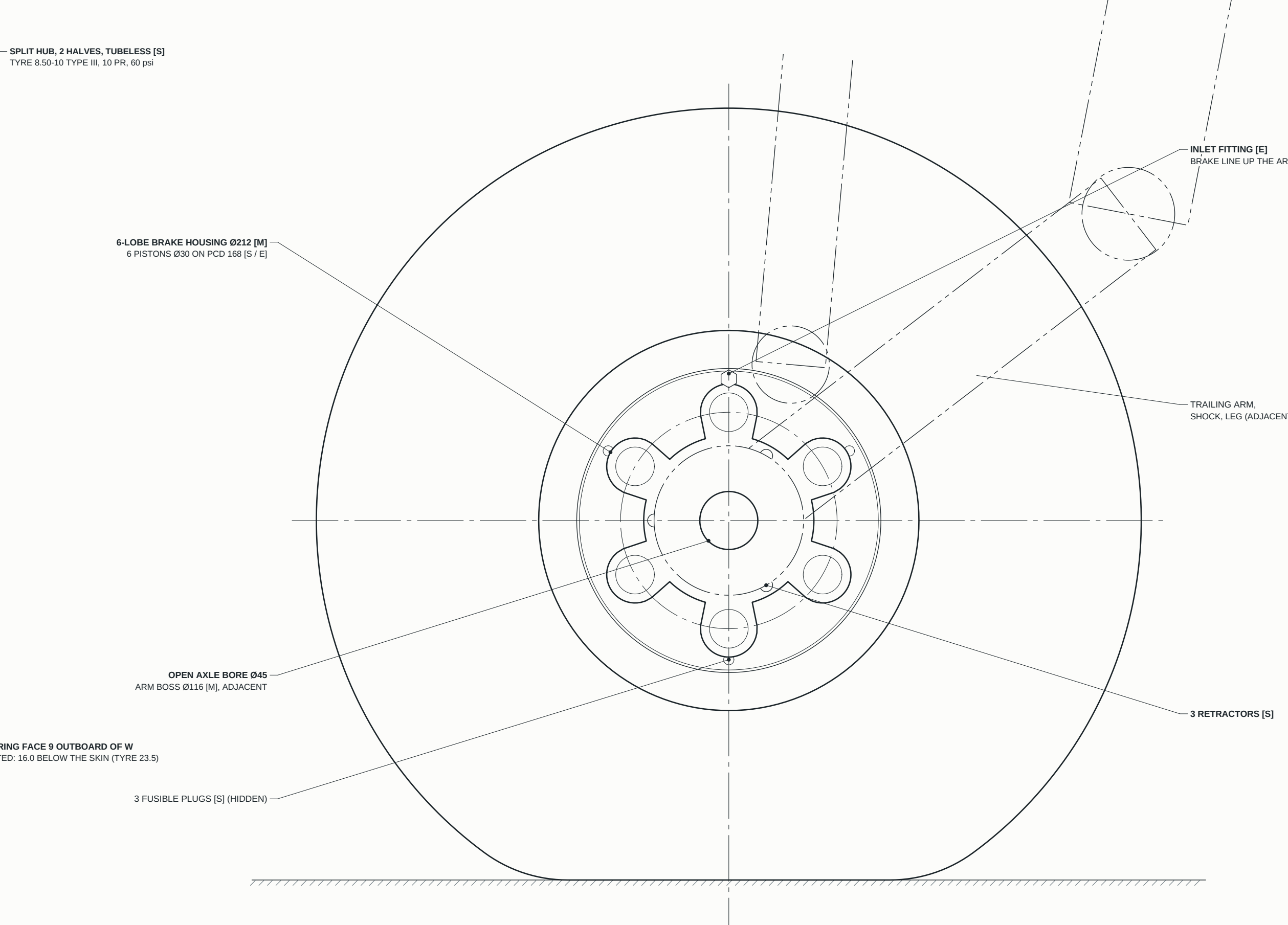
- NOTES
- Drawn from the parameter tables of modelwheels.py (PTable: value + source tag + note) tyre envelopes MAIN_TYRE_ENV, NOSE_TYRE_ENV, sections, rims, fairing, brake, axle, never from the mesh. Axle positions and the gear corner (trailing arm, leg, shock, nose, fork) from modelgear.py, drawn as adjacent parts (phantom). Photos and their measurement: working day in the 3D-ground indications.
 - First-angle projection. Each assembly: SIDE view from port (principal), FRONT view from ahead to its right (upper half in section through the axle), TOP view from above below & for the main wheel also view C from starboard (inboard / brake face) at the far right. Each group's side / front views stand on one ground line.
 - MAIN TYRE - modelled 8.50-18 Type III, CONCEPT DECISION 2026-09-26 tyre makers, parts lists, rear photo methods: TRA OD 627-652, drawn hex OD 648; 3-axis / 1/4-ina drawing 25x55-10 (OD 558), superseated, shown in the accent colour. Axles and static stance kept axia WL 279 = the loaded radius, the tyre flat on the ground over a 250 mm patch (41 mm deflection) with a 6 mm sidewall bulge. Nose tyre loaded hex axia WL 207 photos 200-210, 130 mm patch, 15.5 mm deflection.
 - 3-D model gear MAIN_TYRE = this envelope (R 302.0), the LD-1 leg-door scaffold (R = 12.5 + 332.0) and the round well flange WELL_R1 follow in modelwheels.py main_wheel / nose_wheel revolve / extrude the profiles of this sheet (tyre with the grooves, wheel halves, hub fairing, brake, valve, tie bolts), consistency, 2024.py (LAW) compares the mesh with them.
 - Tyre section: tangent axis - crown fit to the tread half-width, shoulder round, upper sidewall (radius solved: tangent to the shoulder, max width W/2 at 0.50 H), lower sidewall to where the tyre leaves the rim-flange lip round. At the TRA max shoulder with the modelled section is Ø87.6, inside the TRA max Ø87.6.
 - Main hub fairing (PCH) fairings on the outer hubs, paired in the leg-door colour, its face stands 9 mm outboard of the tyre's max width; retracted (wheel steel with the skin) lies 16.0 mm below the skin, inside the tyre 23.5 mm (POW = 1 in, R, Ø44.5).
 - Main brake: six pistons, three rotors, steel friction surfaces, bolted to the axle (PCH 7-4-10), 3 rotors / 2 stators inferred from the overall kit Ø4 pads, 2 stators, 48 rivets). Housing, stack and torque tube stood inside the rebound wheel half (E), one piston tube is shown in section.
 - No markings (owner decision): no sidewall lettering, brand names or placards on wheels or tyres. The tread grooves are geometry (the render-time groove shader is off). Screw, valve and tie-bolt clogging is the drawn pose (the wheels turn); outboard cuts are mirror images. The main leg door (LD-5) is not shown.



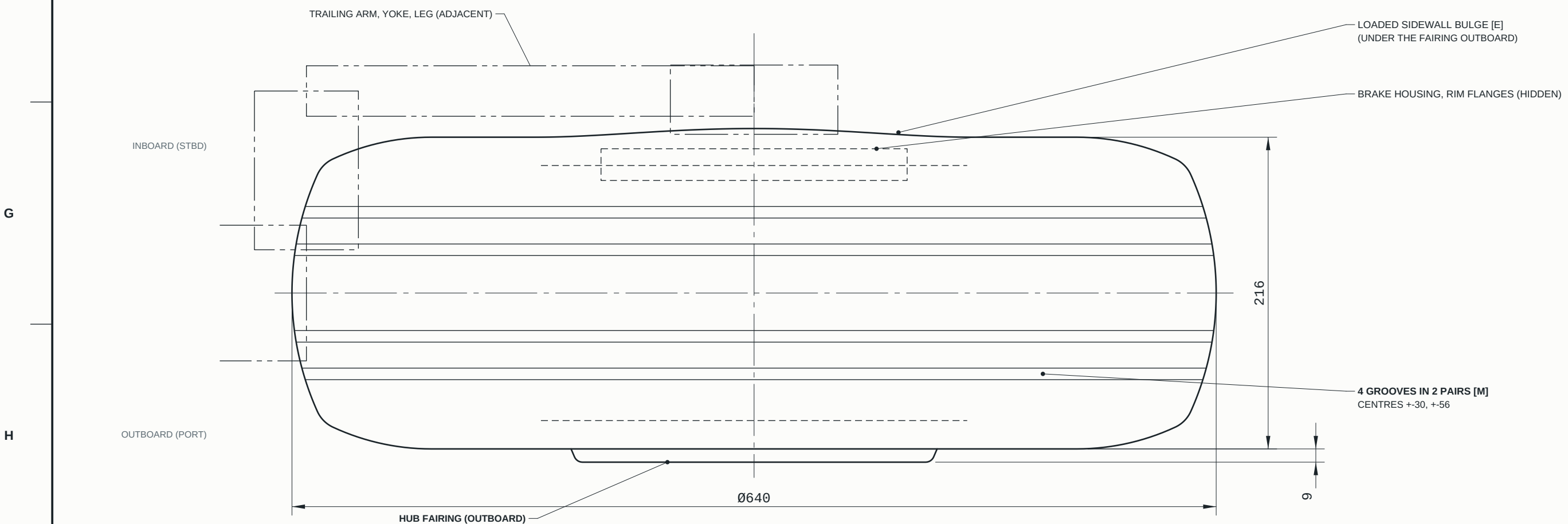
MAIN WHEEL - SIDE VIEW
PORT (UNT SEEN FROM PORT) (OUTBOARD FACE) 1:3



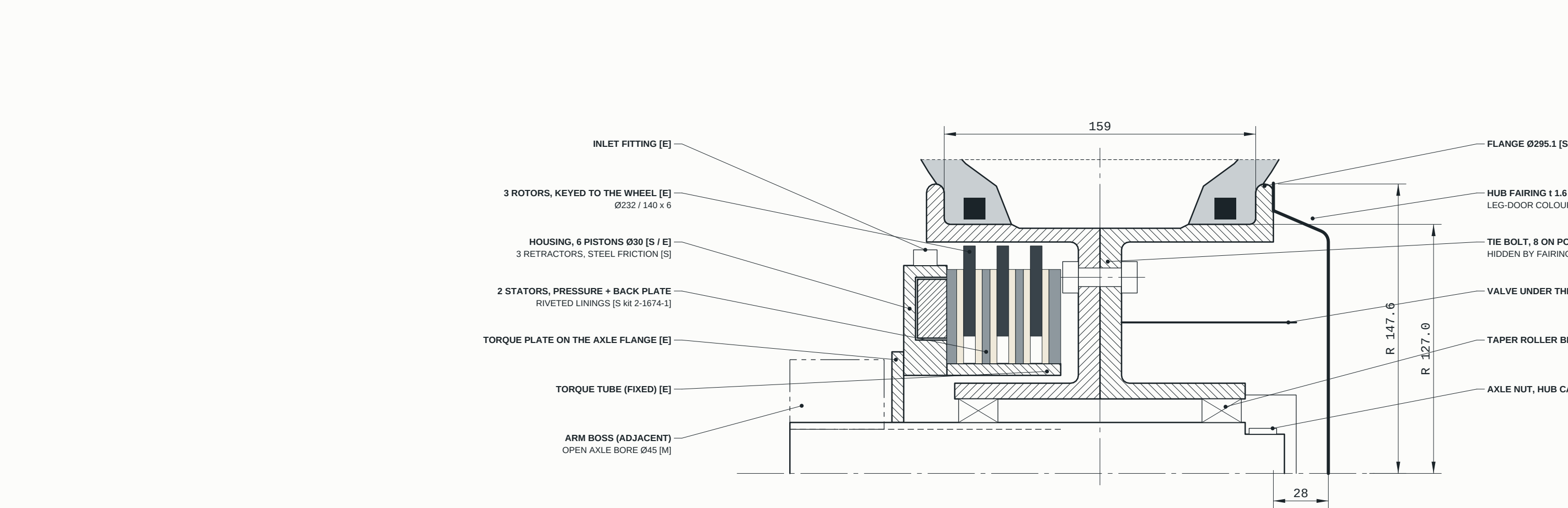
SECTION A-A (HALF)
FRONT VIEW FROM AHEAD, UPPER HALF CUT THROUGH THE AXLE 1:3



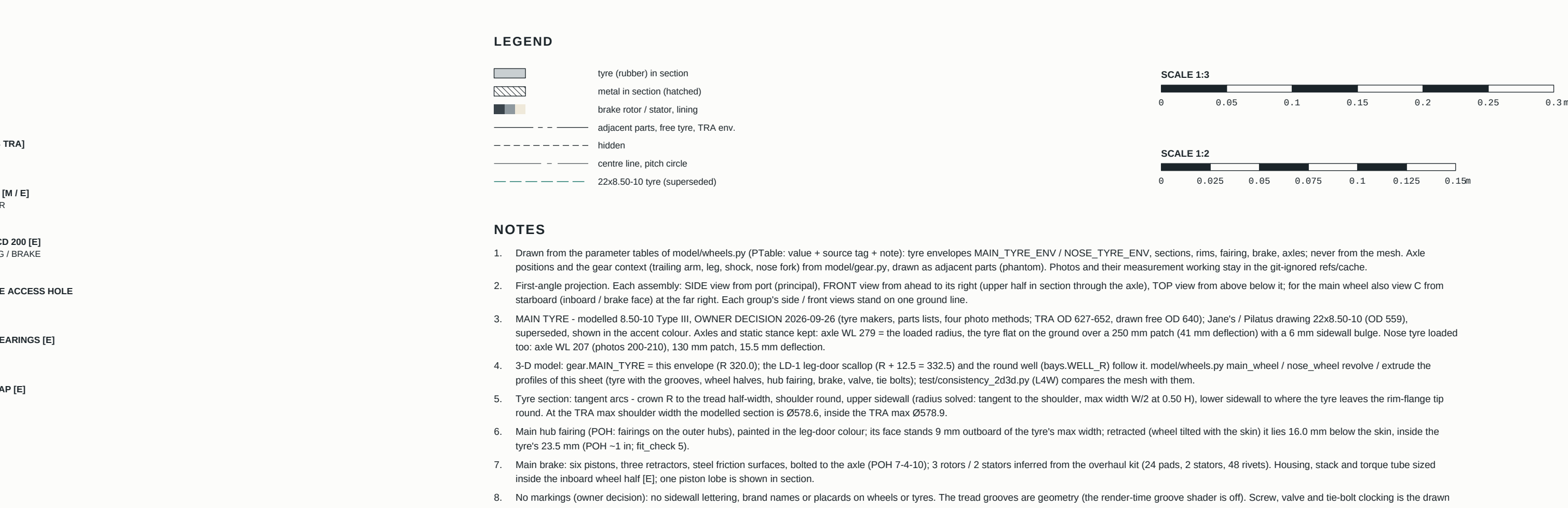
VIEW C - MAIN WHEEL INBOARD FACE
SEEN FROM STARBOARD (BRAKE SIDE), REAR VIEW 1:3



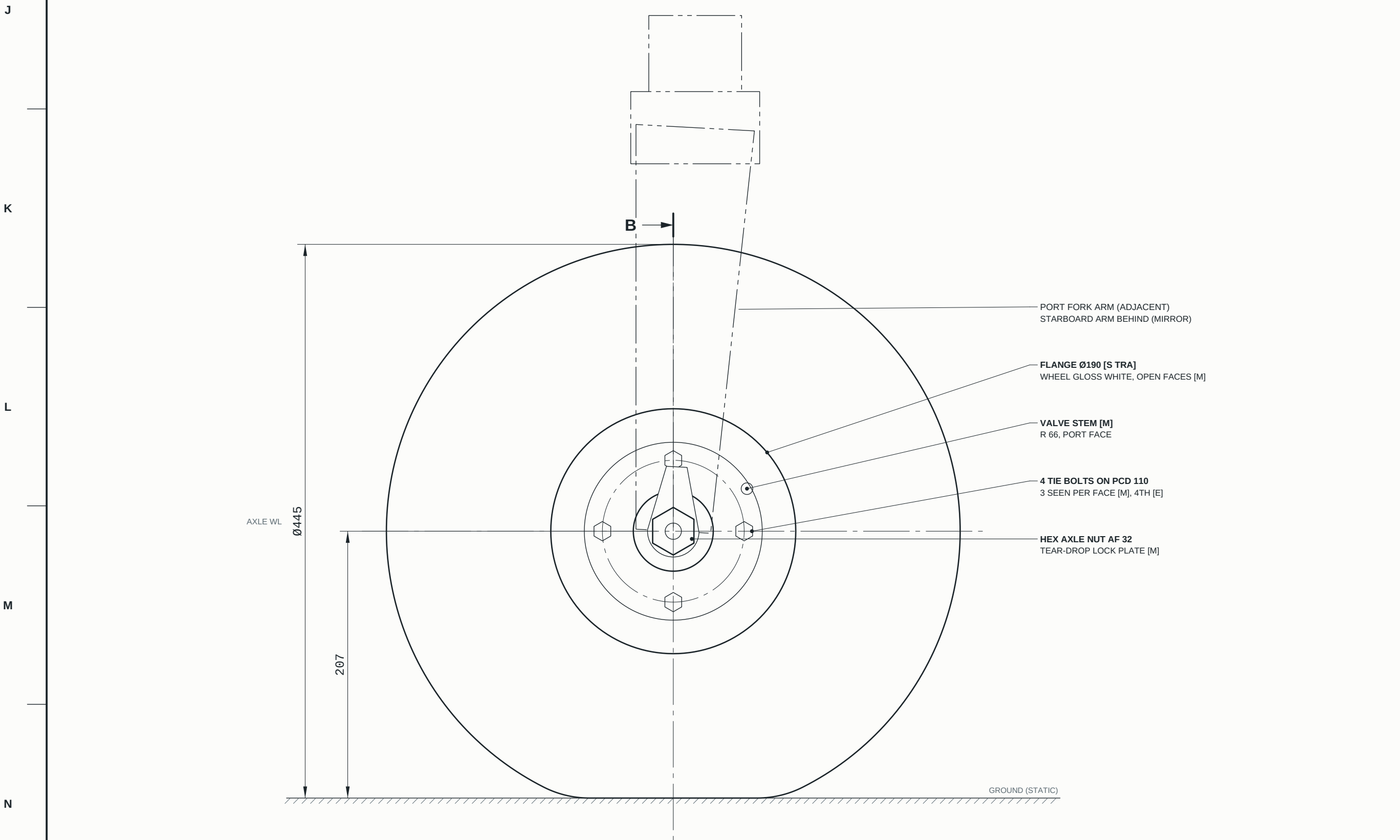
MAIN WHEEL - TOP VIEW
SEEN FROM ABOVE, STARBOARD UP 1:3



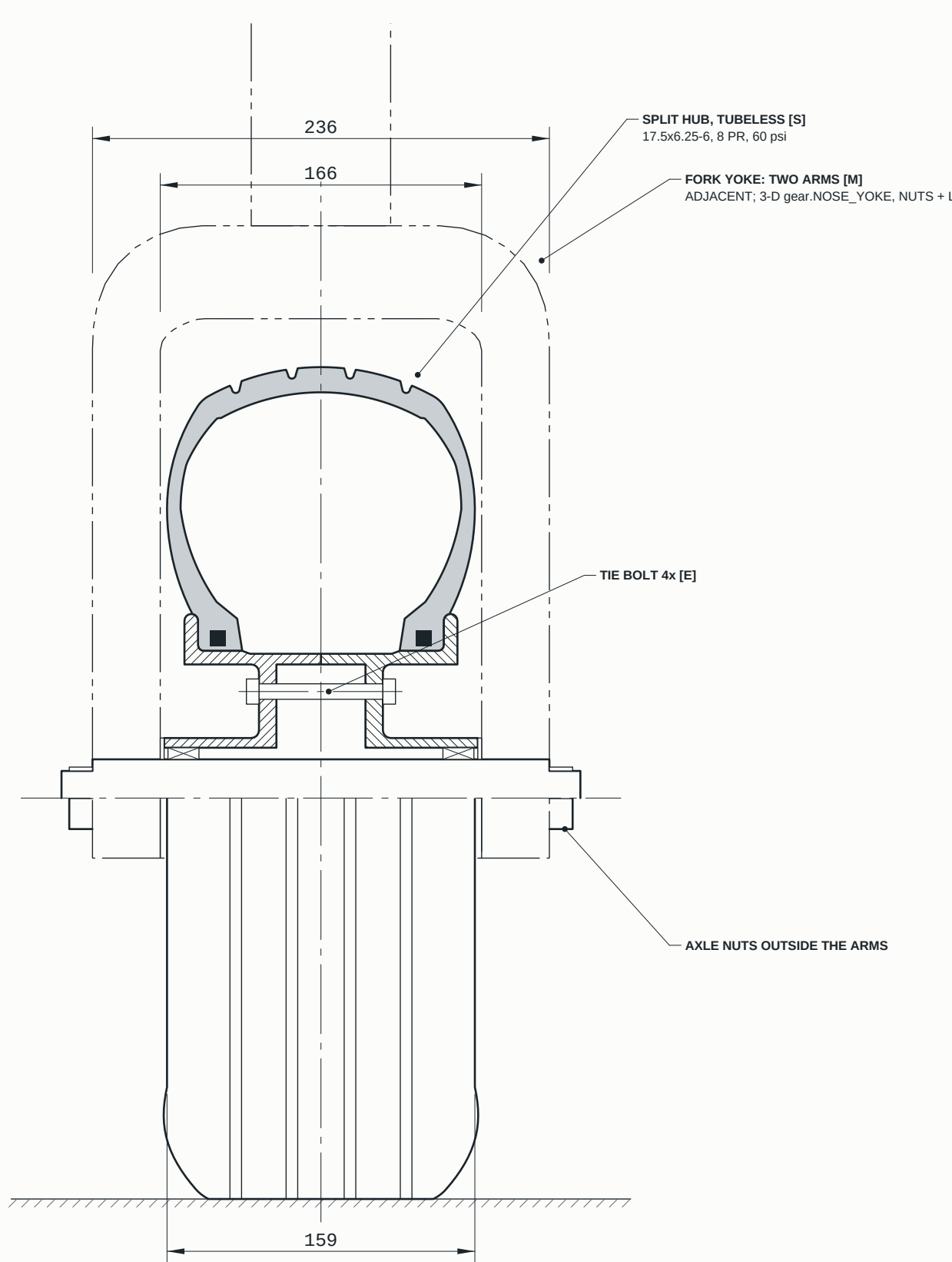
DETAIL H - MAIN HUB AND BRAKE
HALF SECTION A-A, SCALE 1:2



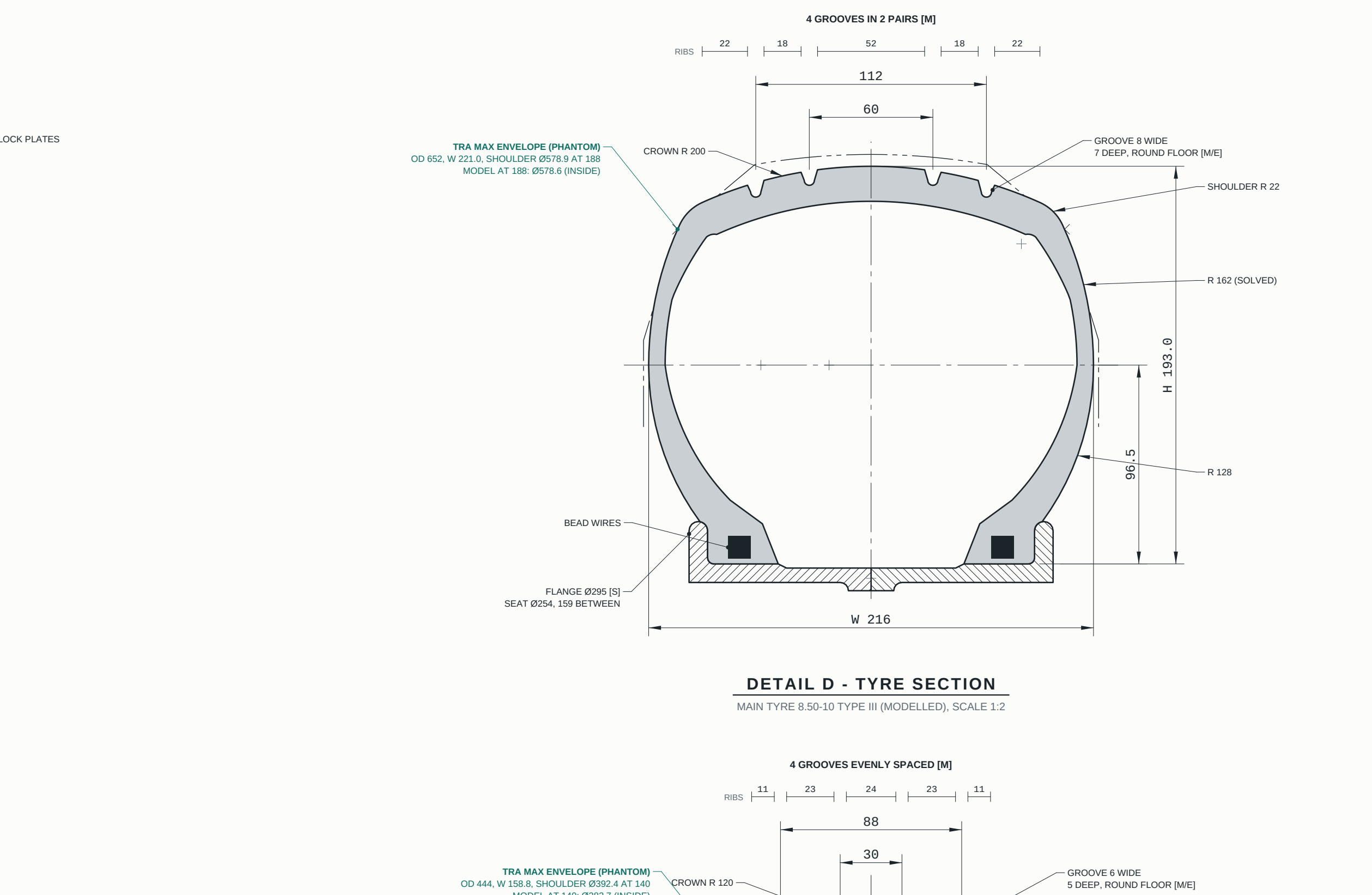
DETAIL D - TYRE SECTION
MAIN TYRE 8.50-18 TYPE III (MODELLED), SCALE 1:2



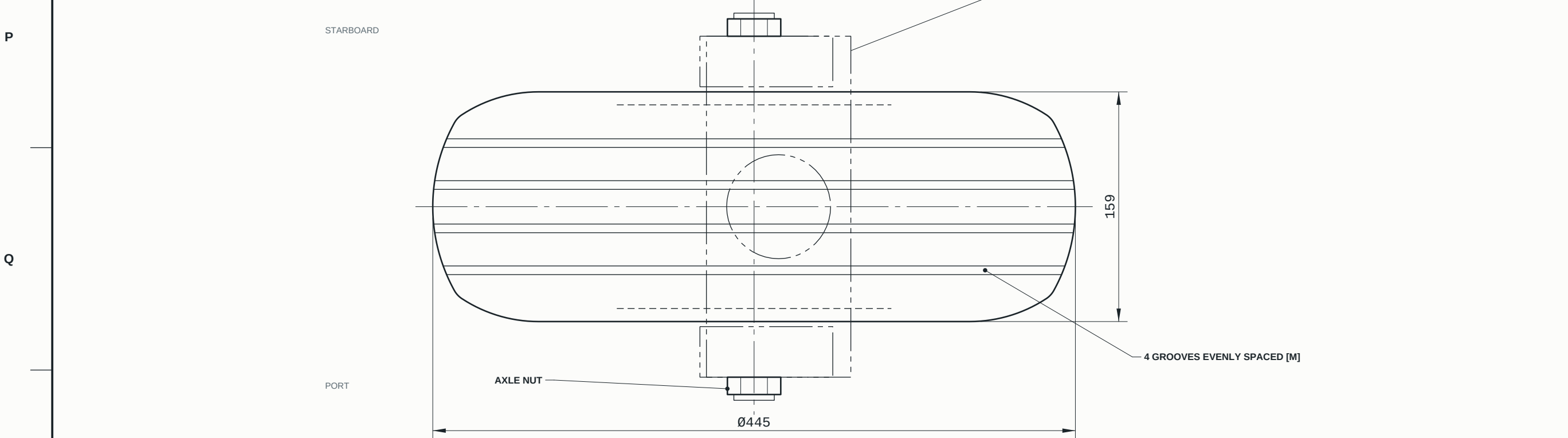
NOSE WHEEL - SIDE VIEW
SEEN FROM PORT 1:3



SECTION B-B (HALF)
FRONT VIEW FROM AHEAD, UPPER HALF CUT 1:3



DETAIL F - TYRE SECTION
NOSE TYRE 17.5x5.25-6, SCALE 1:2



NOSE WHEEL - TOP VIEW
SEEN FROM ABOVE, STARBOARD UP 1:3

PILATUS PC-12 PRO				
WHEELS & TYRES - MAIN AND NOSE WHEEL ASSEMBLIES				FIRST ANGLE PROJECTION
DWG NO	REV	SCALE	SHEET	UNITS
PC12-L4W	A	1:3 / 1:2	A0 · 1 OF 1	mm
DRAWN	Claude - loftkit parametric model			DATE
				2026-09-24
RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION				